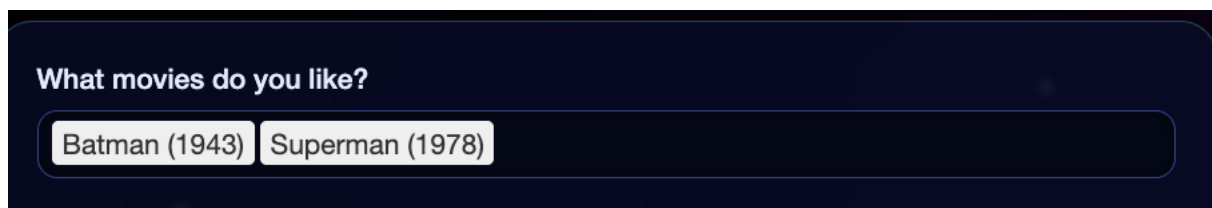


Introduction

The Hidden Gem Movie Recommendation App helps users discover lesser-known movies based on films they already enjoy. By selecting a few favorites and adjusting filters such as confidence, popularity, year range, and genre, the app generates personalized recommendations for “hidden gem” titles that most users have not seen.

User Inputs and How They Work

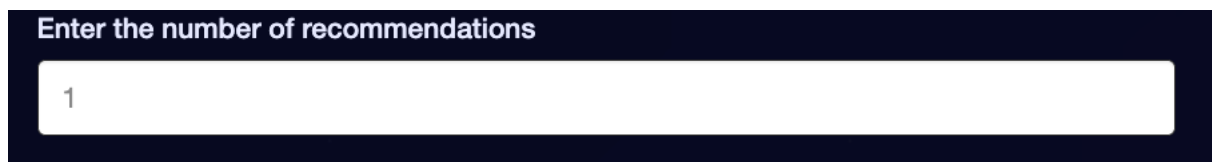
1. Select Movies You Like



The screenshot shows a dark-themed interface with the text "What movies do you like?" in white. Below the text is a horizontal input field containing two grey chips with white text: "Batman (1943)" and "Superman (1978)".

This input allows you to choose one or more movies that you already enjoy. These movies form the basis of your personalized recommendations. Simply start typing a movie title, select it from the dropdown menu, and it will appear as a grey chip. You may select multiple movies before generating results.

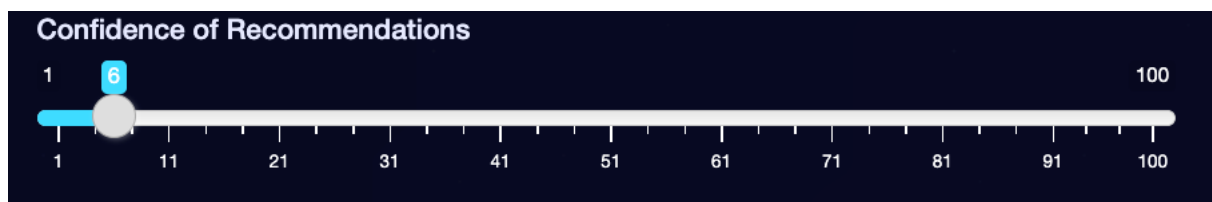
2. Choose the Number of Recommendations



The screenshot shows a dark-themed interface with the text "Enter the number of recommendations" in white. Below the text is a white input field containing the number "1".

This numeric field lets you specify how many movies you want the app to return. Enter any positive integer (for example, 5, 10, or 20) depending on how many hidden gems you'd like to explore.

3. Adjust the Confidence Level



The screenshot shows a dark-themed interface with the text "Confidence of Recommendations" in white. Below the text is a horizontal slider with a light blue track and a grey knob. The track has tick marks and labels from 1 to 100 in increments of 10. The knob is positioned at the value 6, which is highlighted with a blue square.

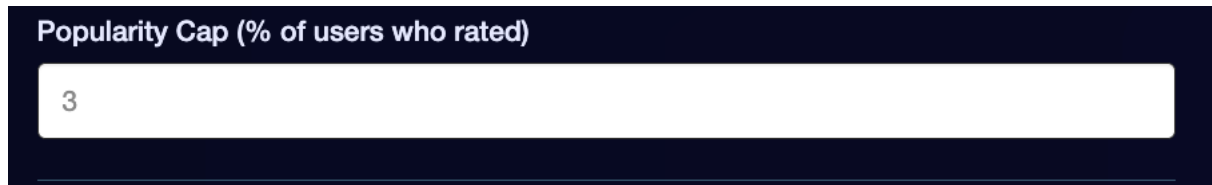
The confidence slider controls how strong the association must be between your selected movies and the recommended titles.

- Lower values (1–20) allow looser, more exploratory matches
- Medium values (20–60) balance flexibility and accuracy

- Higher values (60–100) return only strong matches

A confidence value represents the probability (based on MovieLens data) that a user who enjoyed your selected films would rate the recommended film 4.5 or 5 stars.

4. Set the Popularity Cap

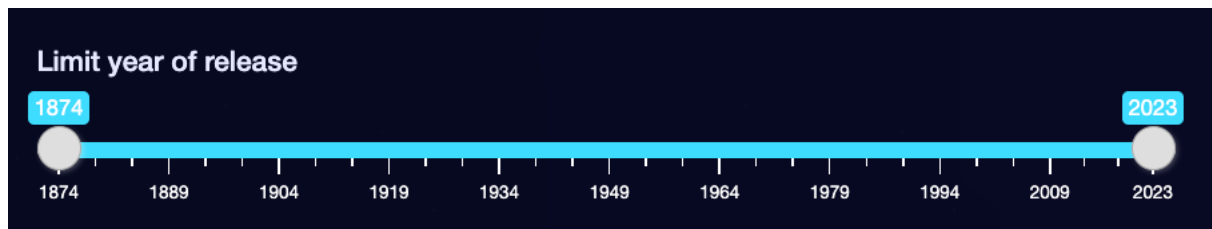
A dark blue rectangular box with the text "Popularity Cap (% of users who rated)" in white. Below the text is a white input field containing the number "3".

This input limits results based on how many users in the dataset have rated the movie.

- Lower percentages (0.1–5%) return extremely obscure films
- Higher percentages (10–30%) allow more well-known titles

Important: This is not the percentage of users who *liked* the film. It is the percentage of MovieLens users who have rated it at all (from 0.5 to 5 stars).

5. Limit the Year Range (Optional)

A dark blue horizontal slider interface. At the top, it says "Limit year of release" in white. Below this is a light blue horizontal bar with two grey circular handles. The left handle is positioned at the year 1874 and the right handle is at 2023. Both years are displayed in white text inside light blue boxes above their respective handles. The bar itself has a series of small white tick marks along its length, with larger white labels for the years 1874, 1889, 1904, 1919, 1934, 1949, 1964, 1979, 1994, 2009, and 2023.

The year range slider lets you filter recommendations by release year. Adjust the left handle for the earliest year and the right handle for the latest year. The default span covers the entire available range of films.

6. Choose Genres to Include (Optional)



Limit to genres (optional)

- ☐ (no genres listed)
- ☐ Action
- ☐ Adventure
- ☐ Animation
- ☐ Children
- ☐ Comedy
- ☐ Crime
- ☐ Documentary
- ☐ Drama
- ☐ Fantasy
- ☐ Film-Noir
- ☐ Horror
- ☐ IMAX
- ☐ Musical
- ☐ Mystery
- ☐ Romance
- ☐ Sci-Fi
- ☐ Thriller
- ☐ War
- ☐ Western

You can restrict recommendations to specific genres such as Action, Comedy, Drama, Thriller, Sci-Fi, or Romance. Select one or multiple genres, or leave all unchecked for no genre filtering.

7. Click “Get Recommendations.”



After adjusting your inputs, scroll down and click Get Recommendations. The app will process your selections and generate a list of hidden-gem movies that match your criteria.

Understanding the Output Table

Movie	WhyRecommended	Confidence	PercentRated	Genre	RatingOutOf5	Year
John Wick: Chapter 3 – Parabellum (2019)	John Wick (2014), Superman (1978)	27.1	2	Action Crime Thriller	3.77	2019
Dredd (2012)	John Wick (2014), Superman (1978)	24	2.78	Action Sci-Fi	3.62	2012
Mission: Impossible - Fallout (2018)	John Wick (2014), Superman (1978)	21.9	1.99	Action Adventure Thriller	3.79	2018

Each recommendation appears in a sortable table. The columns are:

- **Movie**
 - The title and year of the recommended film.
- **Why Recommended**
 - Shows which of your selected movies had the strongest association with this recommendation.
- **Confidence**
 - Indicates how strongly the system believes you will enjoy the movie (higher values reflect stronger patterns in the Movielens data).
- **Percent Rated**
 - Shows how many Movielens users have rated the movie. Lower percentages indicate more obscure films (true “hidden gems”).
- **Genre**
 - The genre(s) associated with the film.
- **RatingOutOf5**
 - The average user rating on a 1–5 scale.
- **Year**
 - The release year of the recommended movie.

Conclusion

By combining your chosen movies with adjustable filters for popularity, confidence, genres, and year range, the Hidden Gem Movie Recommendation App helps you uncover lesser-known films aligned with your personal taste. Insert your screenshots in the locations above, and your technical summary will be complete.